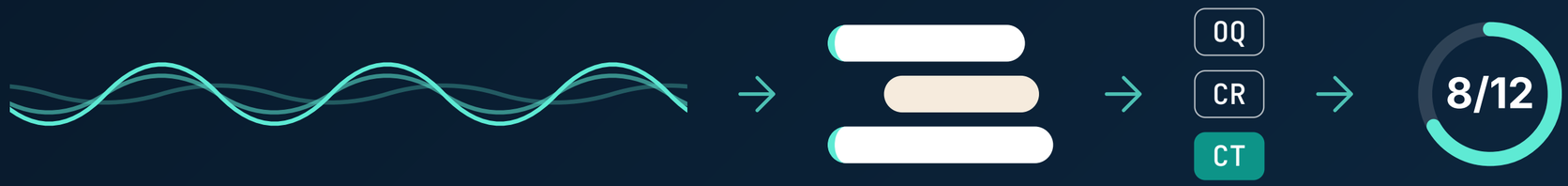
MIRA

MI Review App: expert-grade coaching from any counselling recording



Team **PhoenixAI** · Rahul Nakka · Deependra Shekhawat

Azure Content Understanding · Azure OpenAI in Microsoft Foundry · LangGraph · Streamlit

Same goal. Same three minutes. Opposite outcomes.

SESSION A · ASTHMA

CLINICIAN ▶ 02:58

“If Oscar's that important to you... do you put Oscar higher than your baby, than your wife?”

CLIENT ▶ 03:12

“I don't think you get my situation. I've had this dog for 12 years.”

0% client change talk

SESSION B · VAPING

CLINICIAN ▶ 02:59

“What would you like to do moving forward in terms of vaping?”

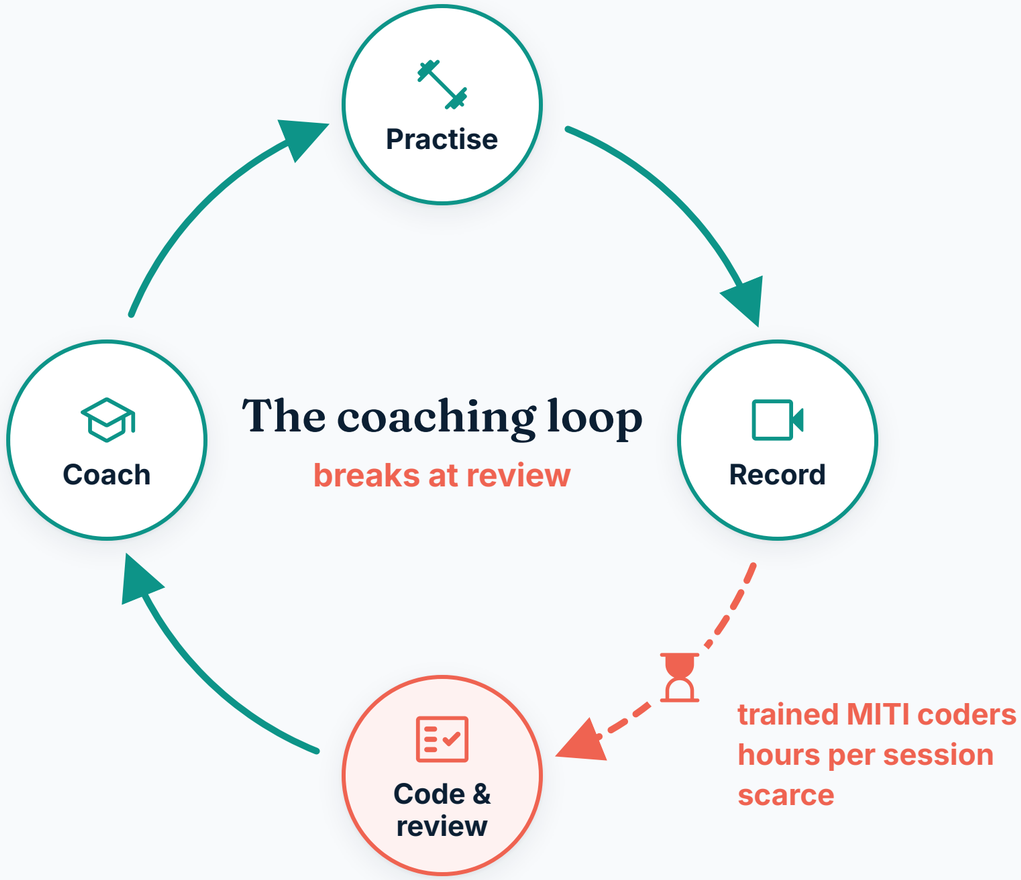
CLIENT ▶ 03:09

“...maybe I could try to, like, cut back a little bit.”

83% client change talk

Public role-play demonstrations (AnnoMI #27; “MI in Action” role-play). Scored by MIRA.

MI skills fade without feedback, and expert feedback doesn't scale.



> “Workshops alone don't make clinicians proficient; feedback and coaching do.”

Miller et al., 2004

> “Training gains fade within months without follow-up coaching.”

Schwalbe et al., 2014

⚠ **Most sessions are never reviewed, so clinicians never learn which words worked.**

Built for three users, around five promises.

Clinician / trainee

“What did I say that worked, and what should I say instead?”

Supervisor / MI trainer

“I can review only a few sessions, and scoring is slow and subjective.”

Programme lead

“Is our MI training changing how we talk to patients?”

OUR FIVE PROMISES

1

Standards-based

MITI 4.2.1

2

Grounded

every claim → timestamp

3

Actionable

what to say instead

4

Fast & affordable

~1 minute, cents

5

Measured

tested vs experts

Good MI: a collaborative stance, moving through four steps.

MI SPIRIT

- **Partnership** — client is the expert
- **Acceptance** — respect autonomy
- **Compassion** — client's interest first
- **Evocation** — draw out their reasons

CLINICIAN ▶ 00:41

“I'm so glad that you came in today... this is a judgment-free zone.”

1
Engaging
build trust

CLINICIAN ▶ 00:41

“Can you tell me about your experience with vaping?”

2
Focusing
agree the topic

CLINICIAN ▶ 02:01

“Can we talk about what some of the pros and cons of vaping are for you?”

3
Evoking
draw out her reasons

CLINICIAN ▶ 02:59

“What would you like to do moving forward?”

4
Planning
her own next step

Miller & Rollnick, *Motivational Interviewing* (2013) · MINT: Motivational Interviewing Network of Trainers · Quotes: Mia's session

Four micro-skills carry the conversation: OARS.

O

Open questions

✗ “Do you vape every day?” CQ

✓ “How does vaping fit into all this for you?” OQ

feeds % open questions

A

Affirmations

“It says a lot about who you are... that you're thinking about the impact.” AF

Name real strengths and efforts.

feeds % MI-adherent

R

CORE SKILL

Reflections

SIMPLE

“Vaping helps you relax.” SR

COMPLEX

“It helps you unwind, and it's starting to cost you on the court, which matters to you.” CR

feeds R:Q and % complex

S

Summaries

“Gather the threads, then move forward.”

Collect what the client said, then transition.

coded as a complex reflection

Experts look for more reflections than questions, and deeper (complex) reflections.

Clients talk themselves into change, or out of it.

← Sustain talk

Change talk →

CLIENT ST
“It helps me relax.”

CLIENT ST
“I don't think I'm addicted.”

Neutral

facts, background, small talk

N

CLIENT CT
“I felt out of breath and super tired.”

CLIENT CT
“Maybe I could try to cut back.”

STATUS QUO — CHANGE

Change talk sounds like: Desire · Ability · Reasons · Need · Commitment · Activation · Taking steps

TRAPS THAT CREATE SUSTAIN TALK

Expert · Question–answer · Premature focus · Labelling / blaming · Persuasion without permission

Confrontation–denial ▶ 02:58

“Do you put Oscar higher than your baby?”

Sal's clinician argued for change, so the client defended the dog.

We score on MITI 4.2.1: six metrics, each with a basic and an advanced bar.

TWO LENSES

Every turn

Behaviour counts: questions, reflections, MI-adherent or not; client change / sustain talk.

Whole session

Global ratings: Empathy and Partnership, 1–5.

Empathy (1–5)

AI-RATED

Supervisor rating

MIA **3.5**

1 — 5

Basic ≥ 3.5 Advanced ≥ 4.0

Collaboration (1–5)

AI-RATED

Supervisor rating

MIA **3.5**

1 — 5

Basic ≥ 3.5 Advanced ≥ 4.0

Reflection : Question

PYTHON

`reflections ÷ questions`

MIA **0.50**

0 — 2.5

Basic ≥ 1.0 Advanced ≥ 2.0

% Open questions

PYTHON

`open ÷ all questions`

MIA **100%**

0 — 100%

Basic > 40% Advanced > 50%

% Complex reflections

PYTHON

`complex ÷ all reflections`

MIA **100%**

0 — 100%

Basic > 40% Advanced > 50%

% MI-adherent

PYTHON

`adherent ÷ (adherent + non-adherent)`

MIA **100%**

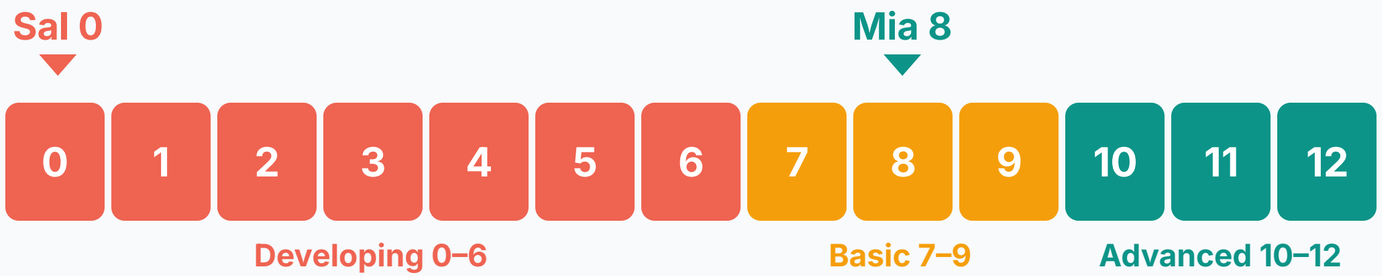
0 — 100%

Basic ≥ 90% Advanced 100%

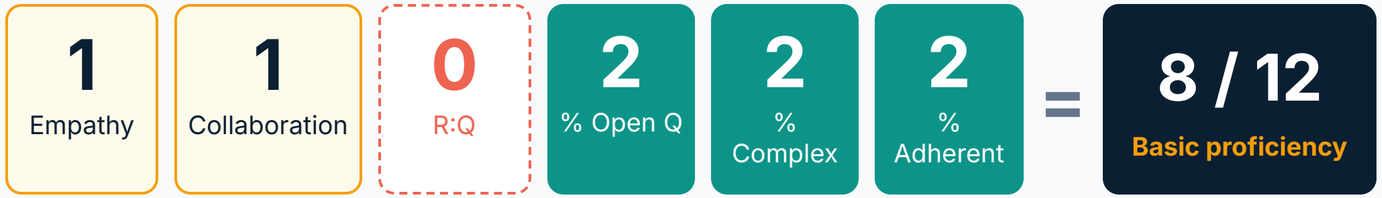
MITI 4.2.1 (Moyers, Manuel & Ernst). Ratings use half-points so 3.5 is reachable. Dots show Mia's session.

Six metrics become one score, and the client's words show if it worked.

MI PROFICIENCY SCORE · 6 METRICS × UP TO 2 POINTS

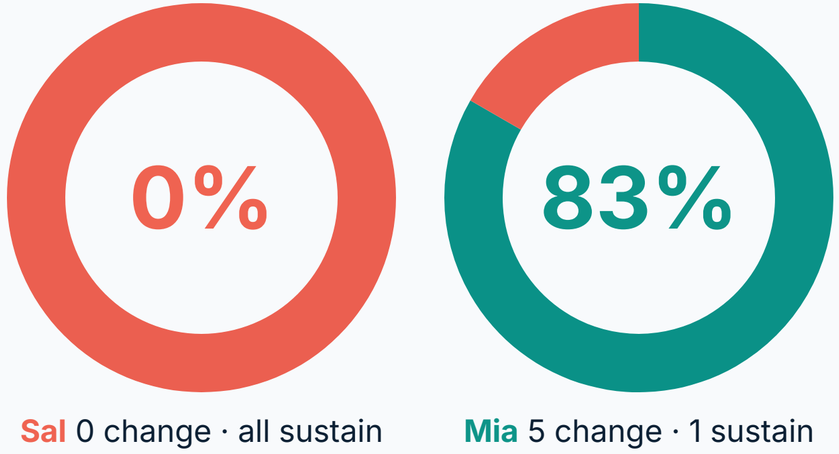


MIA'S BREAKDOWN



CLIENT IMPACT

Change talk ratio = change ÷ (change + sustain)



Client-impact indicator, not part of the 12 points. Neutral turns are excluded.

MIRA: Azure listens, two agents think, Python does the math.

Video / audio in → **MIRA** → Score · ratings · timestamped feedback · coded transcript

LANGGRAPH ORCHESTRATION

1 · LISTEN

Azure Content Understanding

Diarized transcript, ms timestamps, clinician speaker, target behaviour, summary

`turns + timestamps`

2 · CODE

Coder agent · gpt-5.4

MITI / AnnoMI codes for every turn

`codes`

3 · MEASURE

Python rubric engine

Ratios, thresholds, points. No LLM.

`metrics`

4 · SUPERVISE

Supervisor agent · gpt-5.4

Empathy, Collaboration, strengths, traps, better responses

`turn ids`

5 · SHOW

Report + Streamlit

Turn ids → quotes + timestamps; ▶ seeks the video

LLMs judge language; **Python does the math**

Model cites turn ids; **code inserts quote + timestamp**

Strict typed JSON, validated, retried once

AnnoMI: expert-labelled ground truth that powers MIRA four ways.

133
transcripts

9,699
utterances

10
expert annotators

43
topics

SESSION QUALITY



● expert-rated high-quality MI ● low quality (17%)



Only 2 transcripts ever appear in prompts, never in tests (unit-tested).

Wu et al., “Anno-MI: A Dataset of Expert-Annotated Counselling Dialogues”, ICASSP 2022.

The Coder learns from expert examples, not from retraining.

FEW-SHOT LEARNING

Tell zero-shot
Rules only

MIRA ✓

Show few-shot
Rules + ~20 labelled examples

Train fine-tune
Thousands of labels + a training pipeline

PROMPT ANATOMY

Role: expert MI fidelity coder

Coding rules

Clinician & client code definitions

22 AnnoMI examples
~1.4k tokens · auto-selected to cover all 15 labels

cache line · static part cached on repeats

```
[T0] CLINICIAN: …
```

ONE REAL TURN

CLINICIAN — OQ — SEEK — ▶ 02:01

"You're clearly passionate about volleyball and your future. Can we talk about what some of the pros and cons of vaping are for you?"

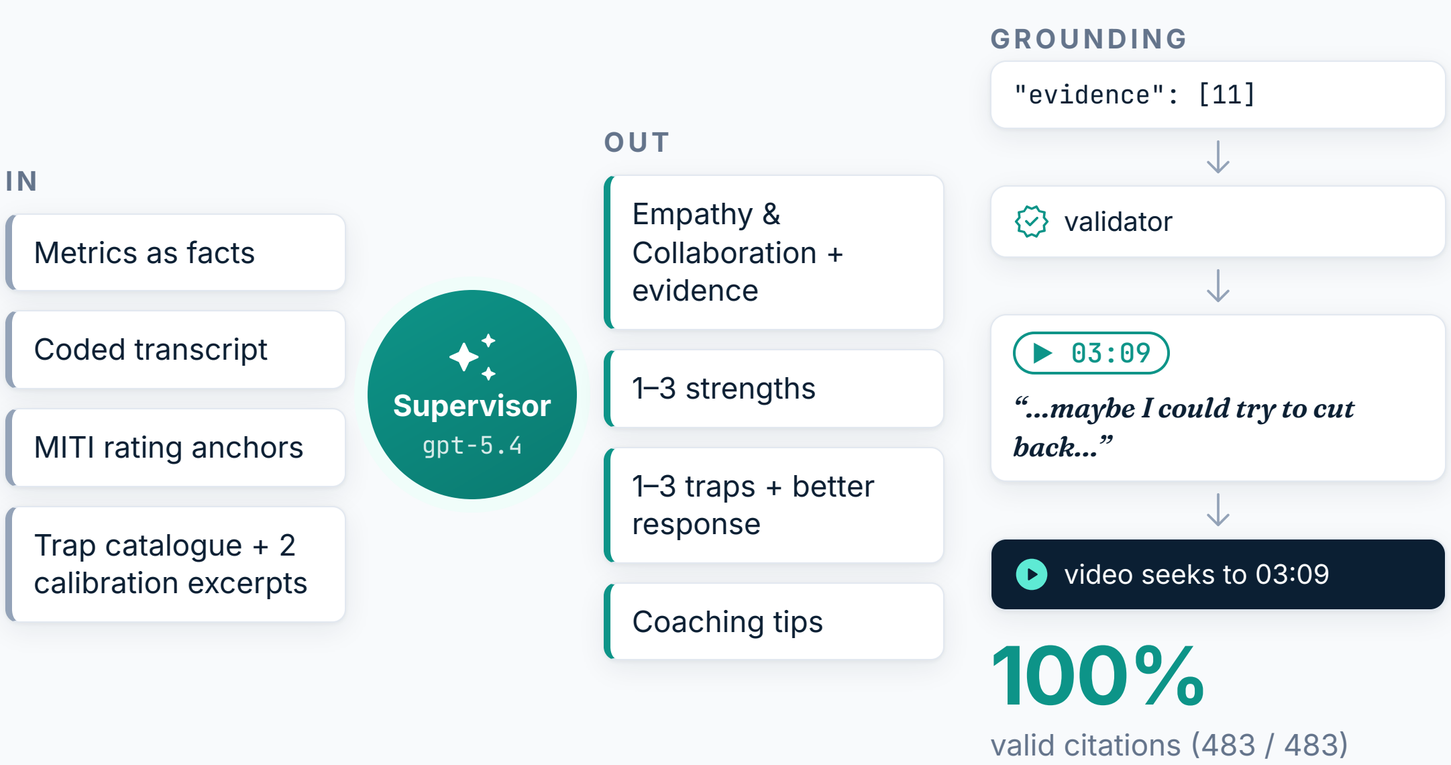
```
{ "turn": 6,
  "rationale": "Affirms values, asks permission,
    then invites pros and cons discussion.",
  "question": "open",
  "mi_behaviour": "seek_collaboration",
  "adherence": "adherent" }
```

CLIENT — CT — ▶ 02:09

"Well, pros, it helps me relax... but I heard it's not super..."

every turn coded exactly once · codes made consistent · strict schema

The Supervisor rates like an MI trainer, and every claim is checkable.



Reflect the change talk before moving to solutions

`missed_change_talk` ▶ 03:09

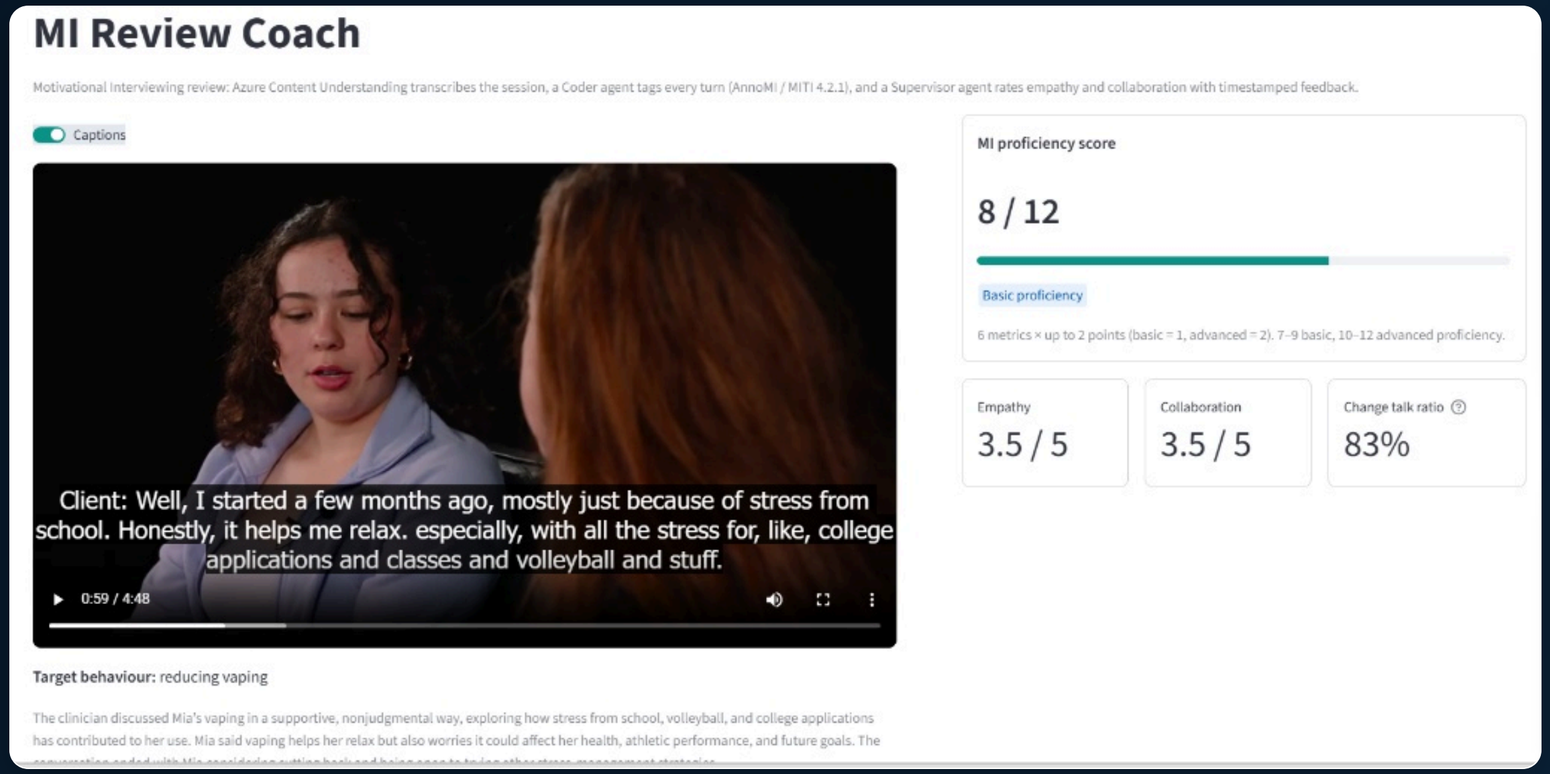
Mia offered strong change talk: she could cut back and needs another way to cope with stress. The clinician moved quickly into offering options.

BETTER RESPONSE

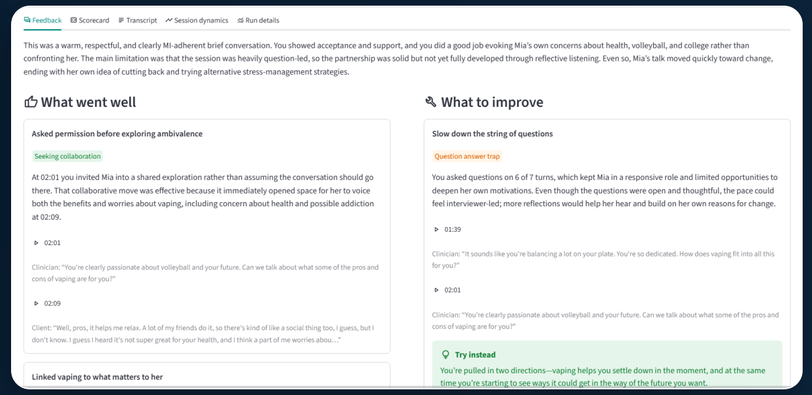
“Cutting back feels more realistic to you right now, and the bigger challenge is finding another way to handle all that stress without leaning on vaping.”

Live demo: MIRA reviews Mia's session.

1. **Analyze** — clip 0:39–3:43, live stages
2. **Overview** — score + 3 KPIs
3. **Feedback** — ▶ seek, better responses
4. **Scorecard** — vs AnnoMI medians
5. **Transcript** — coded, filterable
6. **Dynamics** — run time, tokens



2 Overview · 8 / 12 Basic proficiency, change talk 83%



3 Feedback · timestamped strengths, traps, "try instead"



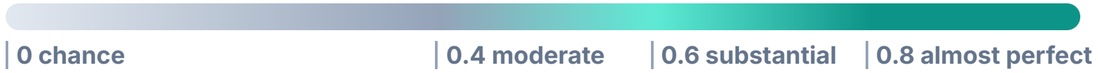
6 Dynamics · change talk overtakes sustain talk

Plan B: the saved report opens instantly, with no Azure call.

How to read our results: four yardsticks.

Cohen's kappa (κ)

Agreement **beyond chance**. A coder answering “neutral” every time scores 64% accuracy but **κ = 0**.



→ **MIRA: κ 0.86 on main clinician behaviour**

Human ceiling

10 experts labelled the same 7 sessions; we compare each with the other 9. **MIRA ≈ ceiling** means as consistent as an expert.



→ **Ceiling: κ 0.81 on main clinician behaviour**

AUC

Pick one good and one poor session at random: how often does MIRA score the good one higher?

0.5 = coin flip → 1.0 = always

→ **MIRA: AUC 0.994 on the proficiency score**

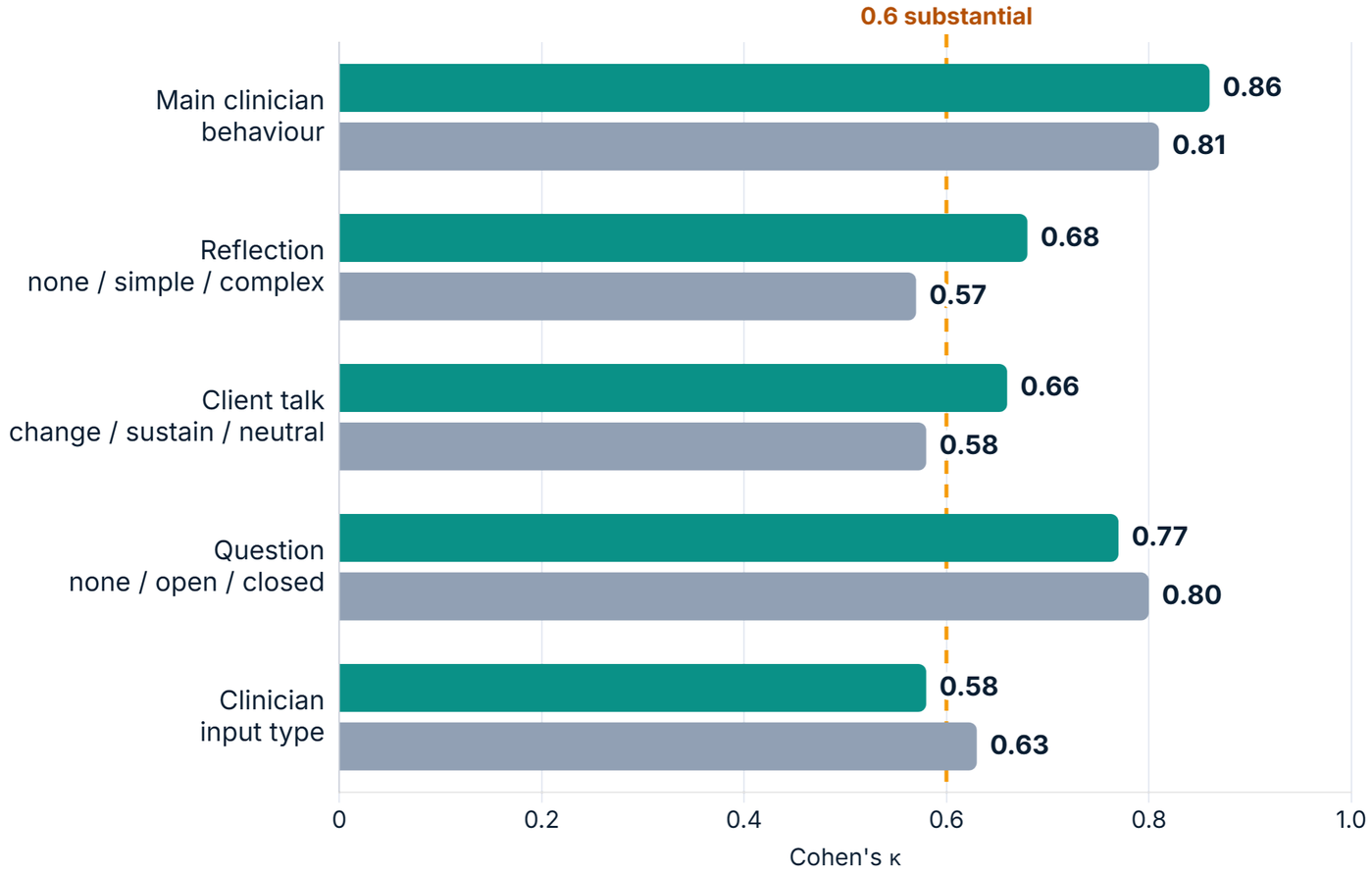
Word error rate (WER)

Share of words transcribed wrong. **10% ≈ 1 word in 10.**

“I felt out of ~~bread~~ **breath** *and super tired at practice.”*

→ **MIRA: 10% WER, 100% speaker roles correct**

The Coder agrees with experts about as well as experts agree with each other.

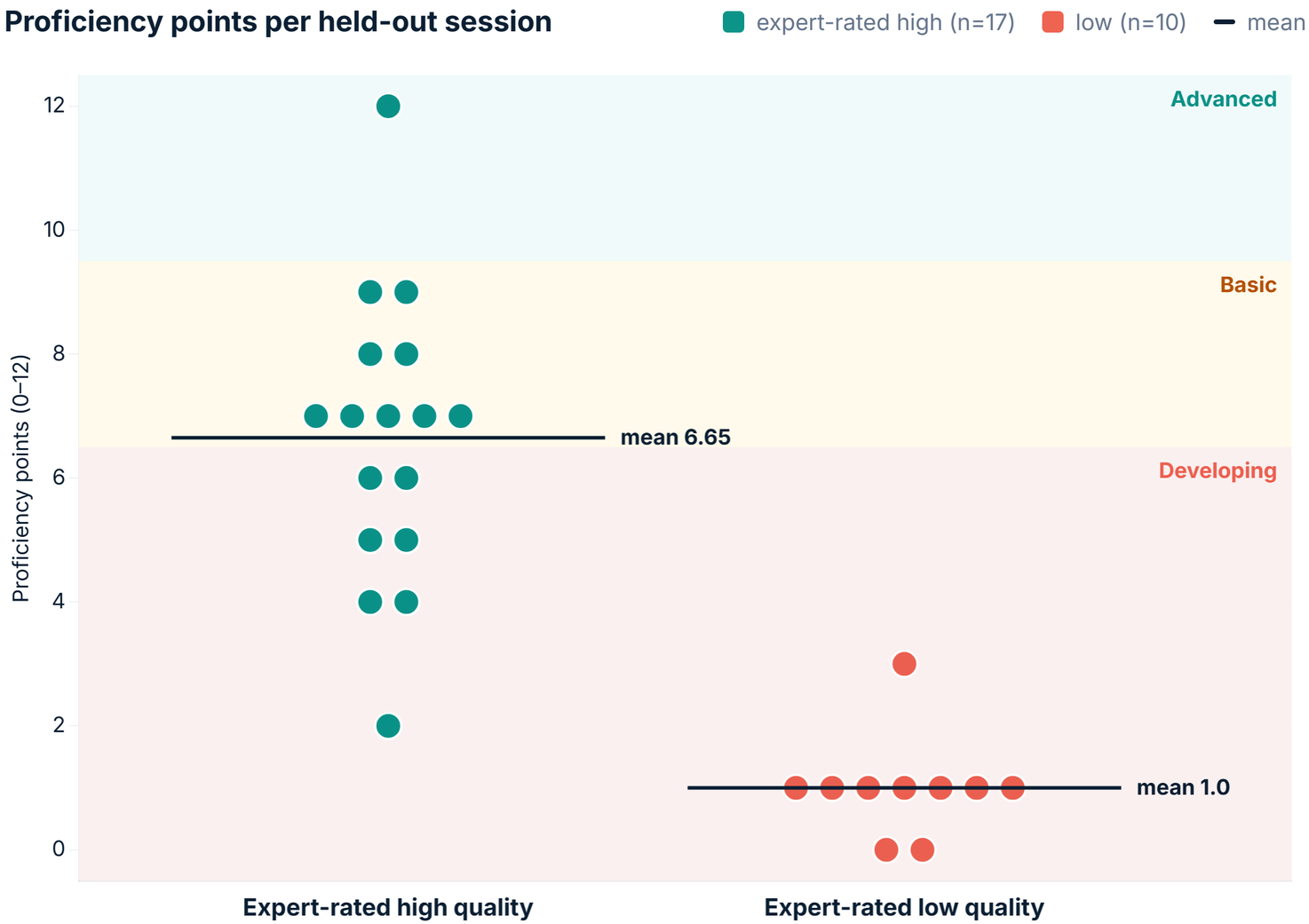


1,276 turns from 27 held-out sessions

Model chosen by data: **gpt-5.4 κ 0.86** vs mini 0.76

One prompt rule (“Yes.” = neutral) lifted client-talk κ **0.47 → 0.61**

MIRA ranks expert-rated good vs poor sessions almost perfectly.



0.994

AUC, proficiency score
(1 mis-ordered pair out of 170)

0.98
Empathy AUC

1.00
Collaboration AUC

TRUST SIGNALS

100% citations valid	**0** score change on re-runs
10% word error rate	**100%** speaker roles correct

MIRA reviews MIRA: what we still need to improve.

WHAT WENT WELL

- **κ ≈ experts**
- **AUC 0.994**
- **100% grounded feedback**

WHAT TO IMPROVE

Reflection depth wobbles

40 of 110 expert-complex reflections coded simple; demo score varied 6–8 across runs.

FIX
Run the Coder 3× and take the majority vote.

Bands are strict

Ranking is right, but 7 of 17 good sessions land in “Developing”.

FIX
Calibrate bands to AnnoMI percentiles.

Audio merges short turns

70% of expert utterances matched (“Mm-hmm” gets merged).

FIX
Word-level timestamps; split backchannels.

Not yet validated

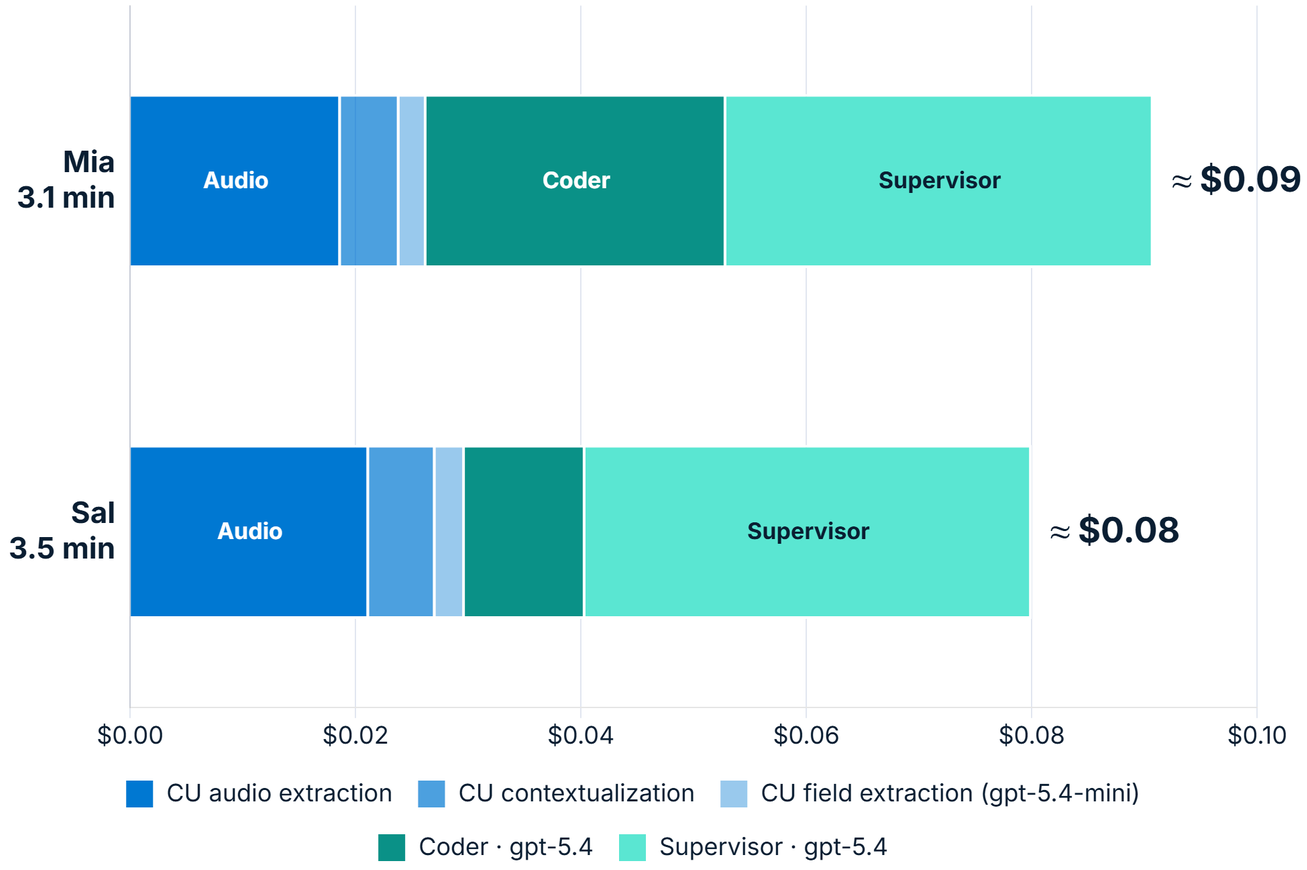
%MI-adherent has no expert labels; text only; prototype (no auth or storage).

FIX
Pilot on real, human-coded sessions.

About 9 cents per 3-minute review, measured, not estimated.

Cost per review (USD), two real runs

~70% model output tokens · ~30% audio



3 min	≈ $0.09		MEASURED
30 min	≈ $0.50	$0.40–0.70	EXTRAPOLATED
60 min	≈ $1.00	$0.80–1.30	EXTRAPOLATED

AT SCALE

100 clinicians × 4 sessions / month (30 min each)

≈ $200 / month

Roughly 1–2 cents per recorded minute.

Azure list prices, Global Standard, pay-as-you-go: CU audio $0.36/h; gpt-5.4 $2.50 in / $15 out per 1M tokens; gpt-5.4-mini $0.75 / $4.50. Token counts from saved reports.

No GPUs, no training: managed Azure services and a tested codebase.

Managed AI

Content Understanding + Azure OpenAI in Foundry; Entra ID auth

Configurable

Models, reasoning effort and analyzer set in `.env`

Quality gates

31 unit tests + 3 eval suites; re-measured on every prompt change

Maintainable

13 Python modules, typed schemas, README recipes

PATH TO PRODUCTION

Prototype

End-to-end pipeline, app, evals, cost telemetry

Pilot

Container Apps / App Service · Entra sign-in · Blob storage with retention · Key Vault · App Insights

Validate on 30–50 consented, human-coded sessions

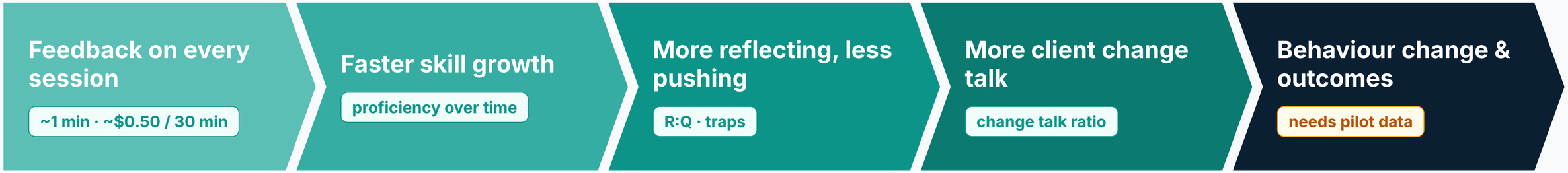
Scale

Team dashboards · LMS / EHR integration · batch reviews (−50%) · more languages

Risks we planned for.

RISK	MITIGATION
Wrong code or score	Evals before every change · grounded evidence · supervisor override · a coaching aid, not certification
Privacy of health conversations	Stays in the organisation's Azure tenant · consent · retention and deletion · role-based access
Feels like surveillance	Feedback goes to the clinician first · managers see aggregates · opt-in
Doesn't generalise (English demo videos)	Pilot on real sessions · recalibrate bands · add locales
Cost or model drift	Token and time telemetry in every report · re-run evals before switching models

Feedback on every session: the lever for better conversations.



MIRA MEASURES ↑

WHO	BEFORE	WITH MIRA
Clinician	Rare, late, general feedback	Every session, ~1 minute, exact moments + what to say
Supervisor	Hours coding a few sessions	Triage “Developing” first; coach instead of count
Programme lead	Training impact invisible	MITI-standard trends by team and metric
Client	Lectured, defensive	Heard; autonomy respected

Next: hear how it's said and see what's shown, measuring behaviour, not emotion.

SAME WORDS

“I suppose I could try.”



change talk

“I suppose I could try.”



masked sustain talk

Text can't tell them apart.

1 Voice from timestamps we already have NO NEW AI

Clinician talk-time share · pause after the client speaks · interruptions · speaking rate

2 Audio-native model on key moments

Warmth, pace, hesitation, each with a timestamp

3 Video via Content Understanding `prebuilt-video`

Gaze, nodding, posture, movement synchrony

WE WILL

- ✓ Measure observable behaviour
- ✓ Get consent from both people
- ✓ Show the clinician first

WE WON'T

- ✗ Infer emotion from faces or voices
- ✗ Score clients
- ✗ Enable surveillance

Evidence: vocal synchrony and empathy (Imel et al., 2014); empathy from speech (Xiao et al., 2015); body-movement synchrony and outcome (Ramseyer & Tschacher, 2011). Microsoft retired facial emotion inference in Azure Face (2022).

Five promises, five receipts.

Standards-based

MITI 4.2.1 + AnnoMI expert labels

Grounded

483 / 483 citations valid

Actionable

a better response for every trap

Fast & affordable

~1 minute, ≈ $0.09 per 3-min review

Measured

κ ≈ experts; AUC 0.994

“...maybe I could try to cut back a little bit.”

Every clinician deserves to know which words got them there.

Thank you. Questions?



`<>` github.com/rahulnk09/Motivational-Interviewing-Review-Agent

Team **PhoenixAI** · Rahul Nakka · Deependra Shekhawat

Azure Content Understanding · Azure OpenAI in Microsoft Foundry · LangGraph · Pydantic · Streamlit